

Conditional Probability

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MATH 241

Lecture #5 — Ch. 2.1-2.2

Reminders!

- Homework #2 is due on **Friday, 9/18** at 11:59pm
 - ▶ Probability problems plus an R problem
- Checkpoint #2 on Thursday at the end of class!
 - ▶ Axioms of probability, combinatorics, conditional probability (e.g. 1.4, 1.6, 2.1-2.2)
- Office hours: **today from 3-4:30pm** in Rocky 106A

Probability in R

We will walk through some R code that will be useful in helping solve some probability problems.

It's Conditional



Learning Objectives

- **Visualize** and **interpret** concepts of conditional probability, through the use of formulas and Bayes' theorem
- **Investigate** the Law of Total Probability, and how it connects to identifying conditional probabilities
- **Calculate** conditional probabilities

Why Conditional Probability?

- When we're given partial information about an event, then we can “condition” on it in calculating probabilities
- As a problem-solving strategy: conditional probabilities can be used to compute probabilities more easily

Formulas to Come

Conditional Probability:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Multiplication Rule:

$$P(A \cap B) = P(B|A) \times P(A)$$

Law of Total Probability:

$$P(B) = \sum_{j=1}^n P(B|A_j)P(A_j)$$

Bayes' Theorem:

$$P(A_j|B) = \frac{P(A_j \cap B)}{P(B)} = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^n P(B|A_i)P(A_i)}$$

Definition of Conditional Probability

Definition 2.2.1 from textbook: If A and B are events with $P(A) > 0$, then the conditional probability of B given A , denoted by $P(B|A)$, is defined as:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

- $P(B)$ is the prior probability
- $P(B|A)$ is the posterior probability

Question: What would change if I wanted $P(A|B)$?

Example #1: Dice Rolls

Suppose I roll two fair dice and am curious about the following events:

- A : at least one dice roll is a 3
- B : sum of die is 8

Find $P(B|A)$.

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Find $P(B|A)$.

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

For $P(A)$, we can count or using combinatorics.

Example #1: Dice Rolls

$$\begin{aligned} P(A) &= \frac{\text{number of times a 3 is shown at least once}}{\text{number of possibilities}} = \frac{11}{36} \\ &= \frac{2 \times \binom{6}{1} - 1}{\binom{n+k-1}{k}} = \frac{2 \times \binom{6}{1} - 1}{\binom{6+2-1}{2}} = \frac{11}{\binom{9}{2}} = \frac{11}{36} \end{aligned}$$

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For $P(A \cap B)$, we can count as well.

$$P(A \cap B) = \frac{\text{number of times a 3 is shown at least once and the sum is 8}}{\text{number of possibilities}} = \frac{2}{36}$$

Example #1: Dice Rolls

Then,

$$\begin{aligned}P(B|A) &= \frac{P(A \cap B)}{P(A)} \\&= \frac{\frac{2}{36}}{\frac{11}{36}} \\&= \frac{2}{11}\end{aligned}$$

Activity #1: Cookies

It's autumn, and it's time to bake some cookies! Denote A as the event of using a recipe for chocolate-chip cookies and B as the event of good cookies being baked. After years of baking, I find that the probabilities are:

- $P(A) = 0.7$
- $P(B) = 0.5$
- $P(A \cap B) = 0.42$

Find the probability of good cookies being baked, given that I did use a recipe. Also find the probability of using a recipe, given that good cookies were baked.

Multiplication Rule

With the definition of conditional probability:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

We can multiply both sides by $P(A)$, which results in the multiplication rule:

$$P(A \cap B) = P(B|A) \times P(A)$$

Interpretation: If A has already happened, then when we look at the probability of B , we're doing so with A in mind. The intersection of A of B is a result of this.

Example #1: Dice Rolls

Suppose I roll two fair dice and have the following events:

- A : at least one dice roll is a 3
- B : sum of die is 8

Find $P(A \cap B)$.

Example #1: Dice Rolls

If I count once again, I can find that:

- $P(A|B) = \frac{2}{5}$
- $P(B) = \frac{5}{36}$

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- $P(A|B) = \frac{2}{5}$
- $P(B) = \frac{5}{36}$

Then,

$$\begin{aligned}P(A \cap B) &= P(A|B) \times P(B) \\&= \frac{2}{5} \times \frac{5}{36} \\&= \frac{2}{36}\end{aligned}$$

Activity #1: Cookies

It's autumn, and it's time to bake some cookies! Denote A as the event of using a recipe for chocolate-chip cookies, but denote C as the event of bad cookies being baked. After years of baking, I find that the probabilities are:

- $P(A) = 0.7$
- $P(C) = 0.3$
- $P(C|A) = 0.1$

Find the probability of using a good recipe with bad cookies being produced.

Multiplication Rule

In general, for the intersection of an arbitrary number n of finite events, if we repeatedly apply the re-arranged definition, we get:

$$P(E_1 \cap E_2 \cap \dots E_n) = P(E_1) \times P(E_2|E_1) \times P(E_3|E_2, E_1) \times \dots \\ \times P(E_n|E_{n-1}, \dots, E_1)$$

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